# Sorting algorithms

Quick sort runs in O(n log n) on average. It picks a pivot, partitions the array, and recurses.

## Comparison

- Fast on average
- Worst case is quadratic

| Algorithm | Best | Worst |
|---|---|---|
| Quick | n log n | n^2 |
| Merge | n log n | n log n |

```
def qs(a):
    return sorted(a)
```